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# Unit - 2(Structural Organisation in Animals and Plants) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

**Anmol Sharma** • Lesson 33 • Feb 11, 2025

# Unit 2 : Structural Organisation of Plants and Animals 200 MCQs

Good evening
audible?
Ready

BIOLOGYLIVE

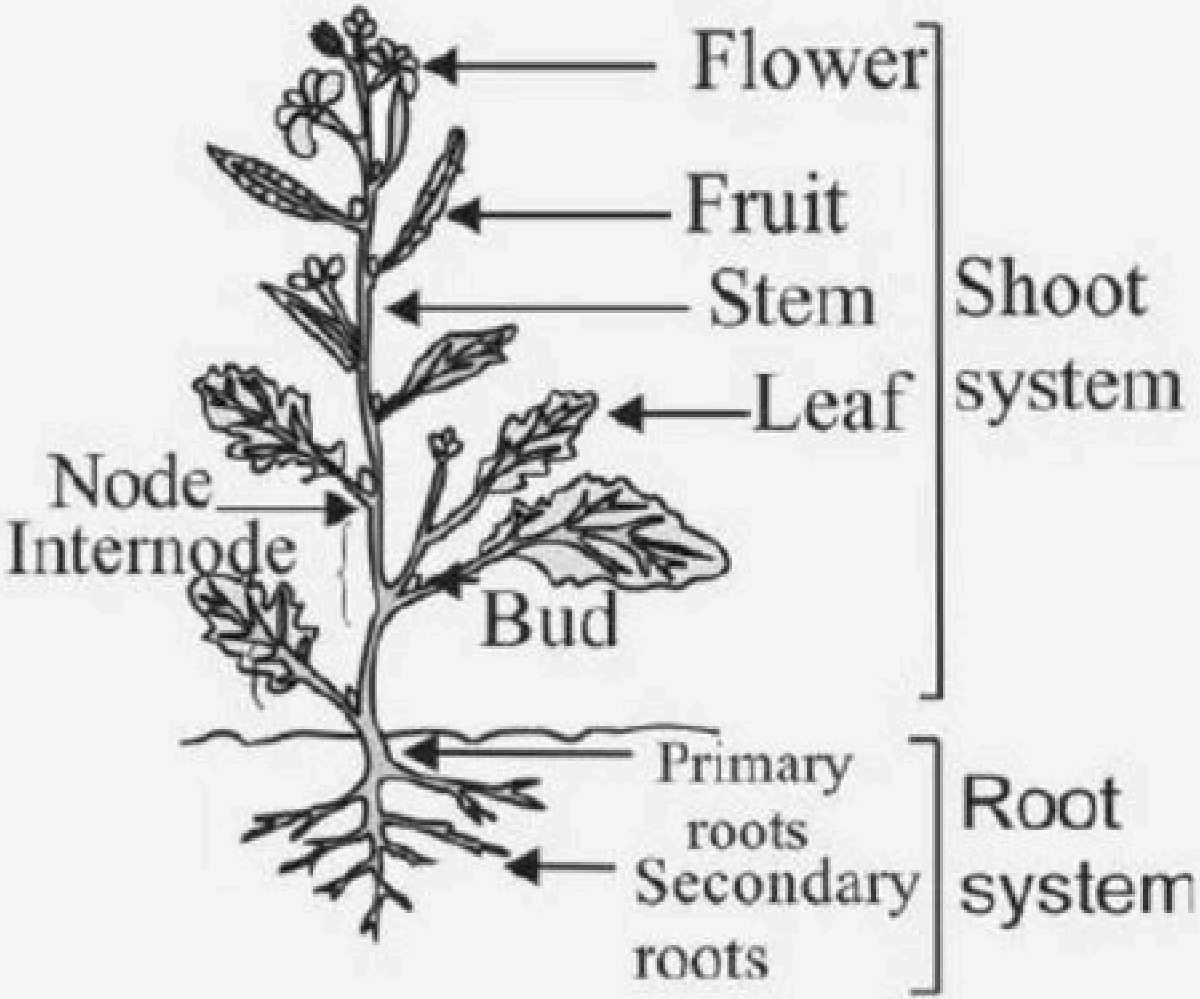
1. In a plant shown in the diagram given below, roots are arised from :-

(1) Base of stem

(2) Plumule

(3) radical X

(4) radicle. ✓



# Title: Unit 2 : Structural Organisation of Plants and Animals



**2. Which of the following is incorrect?**

(1) Roots help in water and mineral absorption from soil

(2) Roots provide a proper anchorage

(3) Roots synthesise plant growth regulators

(4) Roots are always positively geotrophic x

**3. These roots arise from the base of a stem, mark the correct example____** fibrous root

(1) Mustard (2) Datura

(3) Wheat (4) Monstera

**4. Root hairs arise from :-**

(1) Zone of cell division

(2) Zone of cell elongation

(3) Zone of cell maturation

(4) Root cap region

**5. Which of the following region have cells that undergo rapid elongation and enlargement?**

(1) Root cap

(2) Region of elongation

(3) Region of meristematic activity

(4) Region of maturation

**6. A root is adventitious when it is**

(1) Formed from radicle

(2) Swollen

(3) Formed from plumule / other than radicle

(4) Modified for storage

**7. Statement A : The root is covered at the ~~bottom~~ apex by a thimble-like structure called the root cap**

**Statement B : It protects the tender apex of the root as it makes its way through the soil**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**8. The primary root and its branches constitute**

Tap Root System

(1) Fibrous root system

(2) Tap root system

(3) Both (1) and (2)

(4) None of these

**9. Region of meristematic activity is/has :-**

toward = proximal

away = Distal

(1) Large, thick walled dead cells

(2) Distal end of root

(3) Proximal to elongation region

(4) Epidermal cells with root hairs

**10. Label the A, B, C in a given diagram_**

(1) A - Region of elongation

B - Region of meristematic activity

C - Region of maturation.

(2) A - Region of meristematic activity

B - Region of maturation

C- Region of elongation.

(3) A - Region of elongation

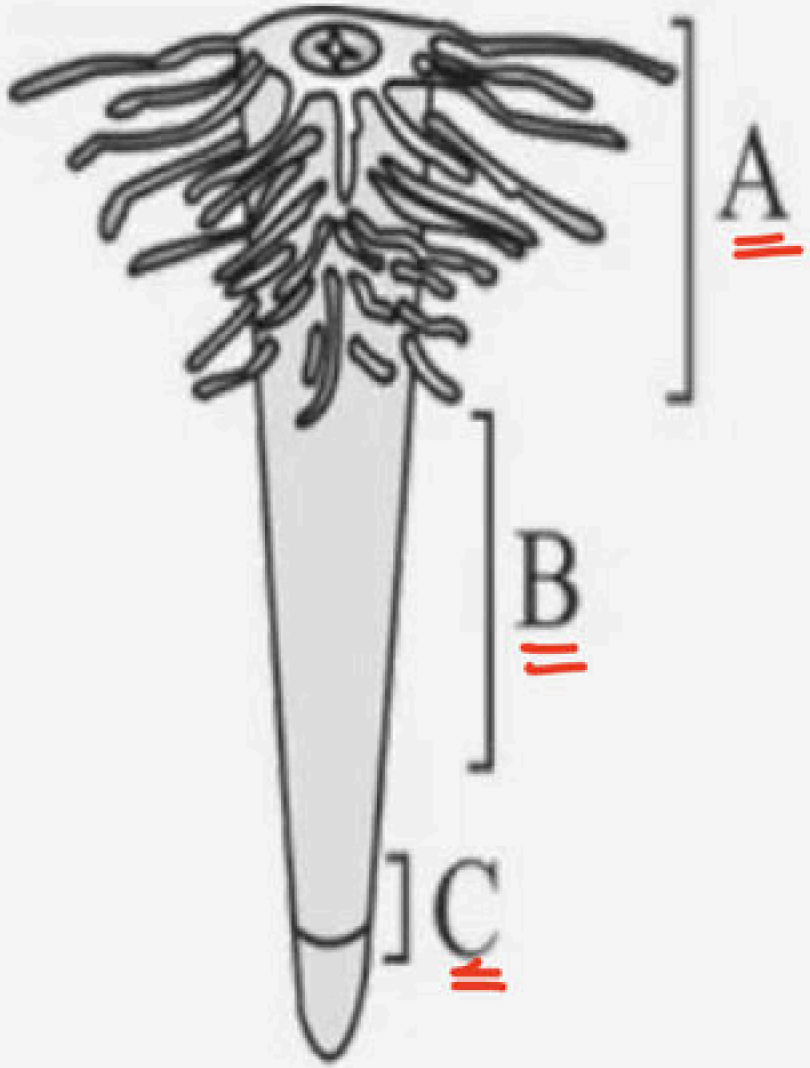
B - Region of maturation

C - Region of meristematic activity.

(4) A - Region of maturation

B - Region of elongation

C - Region of meristematic.



**11. In wheat :-**

(1) Primary root is short lived

(2) Primary root replaced by fibrous roots

(3) Fibrous roots arises from the base of stem

(4) All are correct

**12. Which plant amongst following bears "tap root system"?**

(1) Mustard plant

(2) Wheat plant → Fibrous

(3) Monstera

(4) Grass

**13. The root region is arranged proximal to distal part in the following manner :**

(1) Zone of cell elongation-Zone of cell maturation – Zone of meristematic activity

(2) Zone of cell division - Zone of cell elongation - Zone of cell maturation

(3) Zone of cell maturation-Zone of cell elongation – Zone of meristematic activity

(4) Zone of cell maturation - Zone of cell division - Zone of cell elongation

**14. Which of the following region have cells which are very small, thin-walled and with dense protoplasm?**

100

(1) Root cap

(2) Region of elongation

(3) Region of meristematic activity

(4) Region of maturation

**15. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Fibrous root | P | Turnip |
| B | Tap root | Q | Wheat |
| C | Adventitious root | R | Monstera |

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

**16. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Radicle | P | Fruit |
| B | Plumule | Q | Seed |
| C | Ovary | R | Shoot |
| D | Ovule | S | Root |

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

17 Match the columns I and II, and choose the correct combination with respect to K. Esau.

| Column I | | Column II | |
|---|---|---|---|
| A | Anatomy of seed plants | P | 1954 |
| B | Dr Esau's plant Anatomy | Q | 1957 |
| C | Elected to National Academy of Sciences | R | 1960 |
| D | National Medal of Sciences | S | 1989 |

(1) A-R, B-P, C-Q, D-S
(2) A-P, B-Q, C-R, D-S
(3) A-Q, B-Q, C-P, D-S
(4) A-Q, B-P, C-S, D-R

**18. In monocotyledonous plants, the primary root is replaced by a large number of roots. These roots originate from the base of the stem and constitute the**

(1) Prop roots

(2) Pneumatophores

(3) Tap roots

(4) Fibrous roots

**19. What is the arrangement of root zones starting from "root tip"?**

(1) Cell division, cell maturation, cell enlargement and root cap

(2) Root cap, cell division, cell maturation and cell enlargement

(3) Cell division, cell enlargement, cell maturation, and root cap

(4) Root cap, cell division, cell enlargement and cell maturation

**20. Statement A : In roots, the regionproximal to the region of elongation, is called the region of maturation.** ✓

epidermal.

**Statement B : In roots from the region of maturation some of the ~~endodermal~~ cells form very fine and delicate, thread-like structures called root hairs.** ×

✓ (1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**21. Statement A : In monocotyledonous plants, the primary root is short lived and is replaced by a large number of roots**

**Statement B : These roots originate from the base of the stem and constitute the ~~tap~~ Fibrous root system.** X

✓(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**22. Tap root is descending axis that develops from**

(1) Radicle

(2) Entire seed

(3) Epicotyl

(4) Plumule

**23. Endosperm, a product of double fertilization in angiosperms is absent in the seeds of**

(1) Coconut

(2) Orchids →

(3) Maize

(4) Castor

**24. Monocot characteristics are :**

(1) Fibrous root system, parallel-veined leaves, and one cotyledon

(2) Fibrous root system, parallel-veined leaves, and two cotyledons

(3) Tap root system, parallel venation, and one cotyledon

(4) Fibrous root system, reticulate veined leaves, and one cotyledon.

**25. Which amongst the following options contains mismatched pair?**

(1) Direct elongation of radicle-Primary roots

(2) Monocot -Fibrous roots

(3) Roots arising from the radicle-Prop roots X

(4) Dicot -Tap root system

**26. Functions of root are**

(1) Absorption of water and mineral from soil

(2) Anchoring of plant in soil

(3) Storage of food material

(4) All of these

27. Rearrange the following zones as seen in the root in vertical section and choose the correct option.

A. Root hair zone → 4

B. Zone of meristems → 2

C. Rootcap zone → 1

D. Zone of maturation → 5

E. Zone of elongation → 3

(1) C, B, E, A, D

(2) A, B, C, D, E

(3) D, E, A, C, B

(4) E, D, C, B, A

**28. Statement A : In some plants, like grass, Monstera and the banyan tree, roots arise from the radicle** other **and are called adventitious roots** X

**Statement B : Root system can synthesize plant growth regulators.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**29. Angiosperms are differentiated from "gymnosperms" by the presence of**

(1) Roots and stem

(2) Leaf

(3) Fruit and flowers

(4) All of these

**30. Adventitious roots are found in :-**

(1) Grass

(2) Monstera

(3) Banyan tree

(4) All of these

Plumule

**31. Stem develops from the _____ part of the embryo of a germinating seed.**

(1) Cotyledon

(2) Plumule

(3) Radical

(4) Radicle

**32. The region of a stem called node in siginificant because :-**

(1) It has xylem at the node

(2) Leaves are borne at the node

(3) It acts as organ for storage of food

(4) Apical bud develops at this site.

**33. Which of the following is correct for stem?**

**(a) Its main function is spreading of branches and absorbing nutrition from soil.** x

**(b) It bears only terminal bud not axillary bud.** x

**(c) It bears nodes and internodes.** ✓

**(d) It is developed from the plumule of embryo of a germinating seed.** ✓

(1) b, d     (2) a, c

(3) c, d     (4) a, d

Ascending

**34. The stem is the ____ part of the axis, which bears branches, leaves, flowers and fruits.**

(1) Descending

(2) Radicle

(3) Ascending

(4) Plumule

**35. An underground stem of zaminkand, colocasia, turmeric, etc. act as organ of perennation to_**

(1) Tide over conditions favourable for growth

(2) To give rigidity to the plant

(3) To make plant resistant towards aphids

(4) Tide over conditions unfavourable for growth

# Title: Unit 2 : Structural Organisation of Plants and Animals



**36. Match the following stem modifications given in column I (Stem Modifications) with their examples given in column II (Found in) and select the correct combination from the options given below.**

| Column I | Column II |
|---|---|
| A Underground stem | P Euphorbia |
| B Stem tendril | Q Opuntia |
| C Stem thorns | R Potato |
| D Flattened stem | S Citrus |
| E Fleshy cylindrical stem | T Cucumber |

(1) A-P, B-Q, C-R, D-T, E-S

(2) A-Q, B-R, C-S, D-T, E-P

(3) A-R, B-S, C-T, D-P, E-Q

(4) A-R, B-T, C-S, D-Q, E-P

**37. Plant having "achene" type of fruit is a :-**

(1) Runner - Banana

(2) Runner -Strawberry

(3) Stolon- strawberry x

(4) Stolon - banana

**38. Statement A : In banana, pineapple and Chrysanthemum, the lateral branches originate from the basal and underground portion of the main stem, grow ~~vertically~~ horizontally beneath the soil and then come out obliquely upward giving rise to leafy shoots**

**Statement B : Leaves originate from shoot apical meristems and are arranged in an acropetal order.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

39. In banana, pineapple and Chrysanthemum, the lateral branches originate from

basal mainstem

the __A__ and underground portion of the __B__, grow horizontally beneath the soil

Sucker

and then come out obliquely upward giving rise to leafy shoots called as __C__.

(1) A - apical,B - main root, C - sucker

(2) A - basal, B - main stem, C - Corm

(3) A - apical,B - main root, C - corm

(4) A - basal, B - main stem, C - Sucker

**40. Axillary buds produce all except**

(1) Branch

(2) Flower

(3) Spine

(4) Tendril

**41. Eichhornia plant is propagated vegetatively with the**

help of, :- Stem

(1) Stem

(2) Leaf

(3) Corm

(4) Rhizome

**42) Fill in the blanks:**

**i. When in an inflorescence growth is continuous and**

acropetal

**flowers are borne laterally in a__ 1 __succession.**

**ii. When in an inflorescence growth is not continuous,**

basipetal

**the flowers are borne in a__2__ order.**

(1) 2 - acropetal,1 - acropetal

(2) 2 - basipetal,1 - basipetal

(3) 1 - basipetal,2 - acropetal

(4) 1 - acropetal,2 - basipetal

**43. Select the incorrect statements regarding cymose type of inflorescence__**

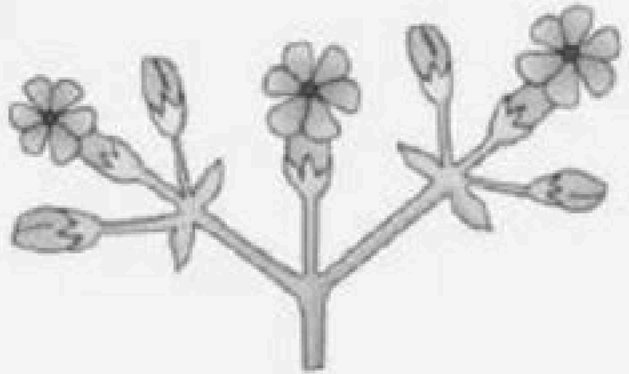
(1) Flowers are born in acropetal order. x

(2) Main axis do not terminate into flower. x

(3) 1 and 2

(4) Flowers are born in basipetal order.

**44. Identify the type of infloresceně mentioned :-**



(1) Cymose

(2) Racemose

(3) Both are correct

(4) None are correct

45. The given diagram represents_ :-

Racemose inflor.

(1) Cymose inflorescence where flowers are borne axially in basipetal manner.

(2) Racemose inflorescence where flowers are borne axially in acropetal succession.

(3) Cymose inflorescence where flowers are borne laterally in acropetal succession.

(4) Racemose inflorescence where flowers are borne laterally in acropetal succession.

**46. The term 'polyadelphous' is related to**

(1) Gynoecium

(2) Androecium

(3) Corolla

(4) Calyx

**47. Select the mismatched option among the following**

(1) Actinomorphic flower- Datura, chilli, potato, mustard

(2) Zygomorphic flower- Cassia, gulmohar, ~~mustard~~

(3) Asymmetric flower- Canna

(4) Both 1 and 2.

**48. Statement A : If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous.**

**Statement B : In epigynous flowers, the margin of thalamus grows upward enclosing the ovary ~~partially~~ completely and getting fused with it.** x

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**49. Identify the correct match :-**

(1) Epiphyllous stamens- ~~potato~~ lily

(2) Pea - monoadelphous

(3) Epipetalous stamens - brinjal

(4) Salvia and mustard - staminode

**50. The mode of arrangement of sepals or petals in floral bud with respect to its other members of the same whorl are known as** :- Aestivation

(1) Phyllotaxy

(2) Inflorescence

(3) Aestivation

(4) Venation

→ calyx

51. X is the outermost whorl of the flower and contains → sepal "Y". Y is green, leaf like and protect the other whorls of the flower .Identify X and Y.

(1) X - Calyx; Y - Sepals

(2) X - Corolla; Y - Petals

(3) X - Gynoecium; Y - Fruit

(4) X - Androecium; Y - Ovary

**52. Based on the position of calyx, corolla and androecium in respect of the ovary on thalamus, the flowers are described as**

(1) Epigynous

(2) Hypogynous

(3) Perigynous

(4) All of these

**53. Select the correct matching about papilionaceous aestivation**

↳ vexillary

(1) Largest petal standard

(2) Two lateral petals - wings

(3) Anterior petals - keel

(4) All are correct

**54. Select the correct matching:**

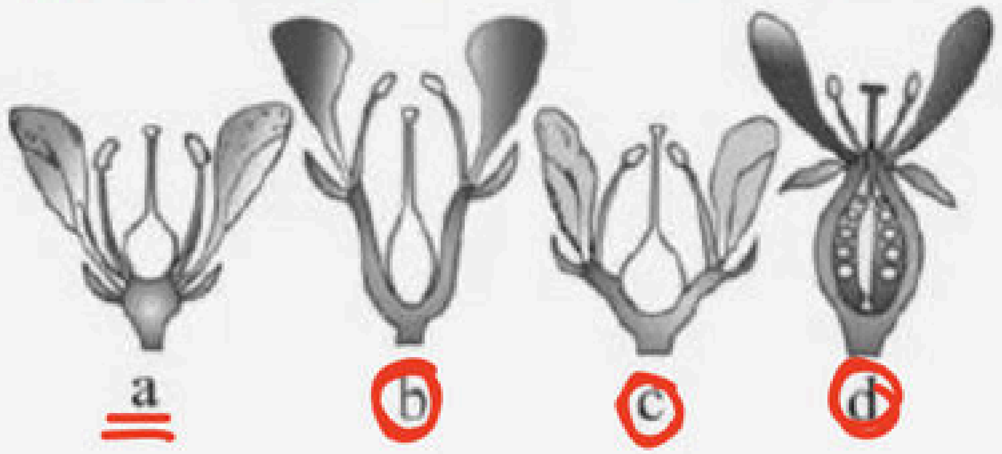
(1) Hypogynous - Superior ovary

(2) Perigynous- Inferior ovary

(3) Epigynous - Half inferior ovary

(4) All of these

**55. Identify the type of ovary depicted in a given diagram_**




(1) a-superior b- half inferior c- half inferior d-inferior

(2) a- inferior b-superior c-half inferior d-superior

(3) a- inferior b-half inferior c-half inferior d-superior

(4) a-inferior b-superior c-inferior d-superior

**56. A typical flower has different kind of whorls _P_ arranged successively on the swollen end of the _Q_ or pedicel called _R_.**

S

(1) P – six Q – stalk R – thalamus

(2) P – four Q – stalk R – receptacle

(3) P – four Q – peduncle R – thalamus

(4) P – six Q – peduncle R – thalamus

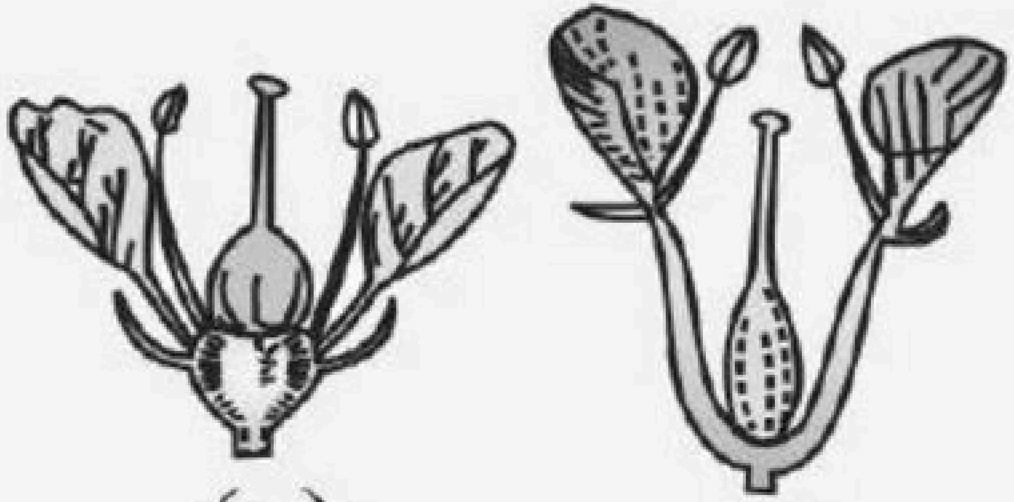
**57. Examples of a and b given in a diagrams respectively are_**

(1) Plum, mustard

(2) Brinjal, rose

(3) Plum, peach

(4) Peach, brinjal



(a)

Superior

Brinjal

(b)

half Inf

rose

**58. Match the columns I and II.**

| Column I | Column II |
|---|---|
| A Monoadelphous | P Citrus |
| B Diadelphous | Q China rose |
| C Polyadelphous | R Pea |

(1) A – R, B – P , C – Q

(2) A – Q, B – R , C – P

(3) A – P, B – R , C – Q

(4) A – Q, B – P , C – R

**59. Statement A : In ~~marginal~~ Parietal placentation, the ovules develop on the inner wall of the ovary oron peripheral part.**

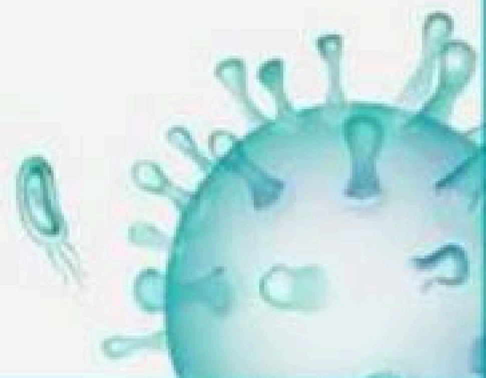
**Statement B : In marginal placentation the placenta forms a ridge along the ventral suture of the ovary.** x

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.



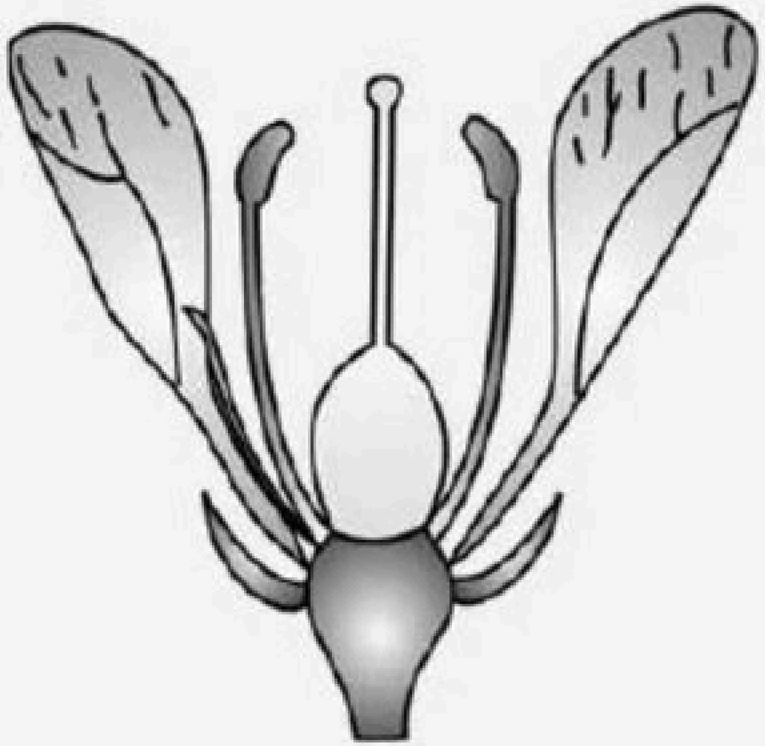
**60. The following diagram depicts the arrangement of floral leaves on thalamus in the flowers of**

(1) Mustard, China rose and brinjal

(2) Plum, rose and peach

(3) Guava, cucumber and the ray florets of sunflower

(4) Rose, China rose and sunflower



Superior

61. Match the columns I and II.

| Column I | | Column II | |
|---|---|---|---|
| A | Actinomorphic symmetry | P | Cassia |
| B | Zygomorphic symmetry | Q | Mustard |
| C | Irregular symmetry | R | Canna |

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

**62. Match Column I (Types of Stamen) with Column II (Example)**
**Choose the correct answer from the options given below:**

| Column I | | Column II | |
|---|---|---|---|
| A | Monoadelphous | P | *Citrus* |
| B | Diadelphous | Q | Pea |
| C | Polyadelphous | R | Lily |
| D | Epiphyllous | S | China-rose |

(1) A – S, B – Q, C – P, D – R
(2) A – S, B – P, C – Q, D – R
(3) A – P, B – Q, C – S, D – R
(4) A – R, B – P, C – S, D – Q

63. Match the columns I and II.

| Column I | | Column II | |
|---|---|---|---|
| A | Inflorescence | P | Polypetalous |
| B | Calyx | Q | Racemose |
| C | Corolla | R | Gamosepalous |
| D | Ovary | S | Parietal |
| E | Placentation | T | Unilocular |

(1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S

64. Match the following placentation types given in Column I (Placentation Types) with their examples given in Column II (Examples) and choose the correct combination from the options given below.

| Column I | | Column II | |
|---|---|---|---|
| A | Basal | P | Dianthus |
| B | Free central | Q | Pea |
| C | Parietal | R | Lemon |
| D | Axile | S | Marigold |
| E | Marginal | T | Argemone |

(1) A-P, B-Q, C-R, D-S, E-T

(2) A-Q, B-R, C-S, D-T, E-P

(3) A-S, B-P, C-T, D-R, E-Q

(4) A-S, B-R, C-T, D-P, E-Q

## 65. Match the columns I and II.

| Column I | | Column II | |
|---|---|---|---|
| A | Variation in length of filaments within flower | P | Lotus |
| B | Apocarpous | Q | Salvia |
| C | Syncarpous | R | Tomato |

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

66. Match the columns I and II, where column I represents type of placentation and column II represents it's example.

| Column I | | Column II | |
|---|---|---|---|
| A | Marginal | P | Argemone |
| B | Axile | Q | Pea |
| C | Parietal | R | Lemon |
| D | Free Central | S | Marigold |
| E | Basal | T | Primrose |

(1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S

**67. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Apocarpous | P | Sesbania |
| B | Syncarpous | Q | Lily |
| C | Diadelphous | R | Mustard |
| D | Epiphyllous androecium | S | Rose |

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

**68. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Mustard | P | Inferior ovary |
| B | Rose | Q | Superior ovary |
| C | Cucumber | R | Half inferior ovary |
| D | Sterile stamen | S | Petals free |
| E | Polypetalous | T | Staminode |

(1) A – R, B – P, C – Q, D – T, E – S

(2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

(4) A – Q, B – T, C – P, D – R, E – S

69. Match the columns I and II.

| Column I | | Column II | |
|---|---|---|---|
| A | Valvate | P | Gulmohar |
| B | Twisted | Q | Calotropis |
| C | Imbricate | R | Cotton |
| D | Vexillary | S | Gamosepalous |
| E | United sepals | T | Trifolium |

(1) A – R, B – P, C – Q, D – T, E – S

(2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

(4) A – Q, B – T, C – P, D – R, E – S

**70. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Mesocarp | P | Outer thin layer |
| B | Endocarp | Q | Middle fleshy/fibrous layer |
| C | Epicarp | R | Inner stony layer |

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

71. Match Column I with Column II

Choose the correct answer from the options given below:

| Column I | | Column II | |
|---|---|---|---|
| A | Rose | P | Twisted aestivation |
| B | Pea | Q | Perigynous flower |
| C | Cotton | R | Drupe |
| D | Mango | S | Marginal placentation |

(1) A – Q, B – S, C – P, D – R

(2) A – P, B – Q, C – R, D – S

(3) A – S, B – R, C – Q, D – P

(4) A – Q, B – R, C – S, D – P

**72. Bone forming cells are**

(1) Osteoclasts

(2) Chondroclasts

(3) Osteoblasts

(4) Chondroblasts

**73. Goblet cells secrete** :- Mucous

(1) Serum

(2) Mucous

(3) Sebum

(4) Milk

74. Match Column - I with Column - II.

Choose the correct answer from the options given below:

| Column I | | Column II | |
|---|---|---|---|
| A | Bronchioles | P | Dense Regular Connective Tissue |
| B | Goblet cell | Q | Loose Connective Tissue |
| C | Tendons | R | Glandular Tissue |
| D | Adipose Tissue | S | Ciliated Epithelium |

(1) A - P, B - Q, C - R, D - S

(2) A - Q, B - P, C - S, D - R

(3) A - R, B - S, C - Q, D - P

(4) A - S, B - R, C - P, D - Q

**75. Which of the following tissue has a free surface, which faces either a body fluid or the outside environment?**

(1) Muscular tissue

(2) Epithelial tissue

(3) Neural tissue

(4) Connective tissue

**76. Gap junctions:-**

(1) Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells

(2) For rapid transfer of ions, small molecules and sometimes big molecules

(3) Provide a synctium

(4) Both 1 and 2

**77. Identify the correct option by reading the correct statement :-**

**I. Striated muscle fibres are bundled together in a parallel fashion.**

**II. Neuroglial cells protect and support the neurons.**

**III. Biceps are i~~nvoluntar~~y and striated.** X

**IV. Skeletal muscle tissue is closely attached to skeletal bones.**

(1) Statement I and IV are correct

(2) Statement I, II and IV are correct

(3) Statement I, II and III are incorrect

(4) Statement I, III and IV are incorrect

**78. Compound squamous epithelium occurs in :-**

(1) Dry surface of the skin

(2) The moist surface of buccal cavity

(3) Inner lining of ducts of salivary glands and of pancreatic ducts

(4) All of them

**79. Match Column- I with Column- II.**

**Choose the correct answer from the options give below:**

| Column I | | Column II | |
|---|---|---|---|
| A | Mast cells | P | Ciliated epithelium |
| B | Inner surface of bronchiole | Q | Areolar connective tissue |
| C | Blood | R | Cuboidal epithelium |
| D | Tubular parts of nephron | S | Specialised connective tissue |

(1) A-P, B-Q, C-S, D-R

(2) A-Q, B-R, C-P, D-S

(3) A-Q, B-P, C-S, D-R

(4) A-R, B-S, C-Q, D-P

## 80. Match the columns I and II

| Column I | | Column II | |
|---|---|---|---|
| A | Adipose tissue | P | Packaging tissue |
| B | Tendons | Q | Connects bone to bone |
| C | Areolar tissue | R | Connects muscle to bone |
| D | Ligament | S | Stores fat |

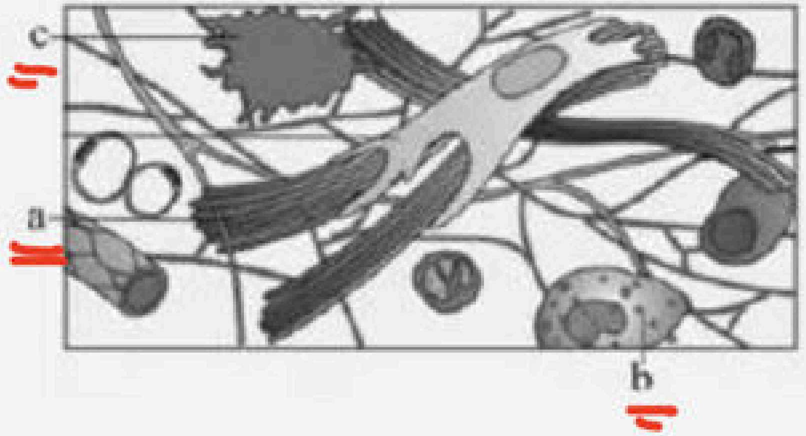
(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

**81. Identify the structures marked as a, b, c**




(1) a-Collagen fibres, b-Mast cells, c-Macrophages

(2) a-Collagen fibres, b-Fibroblast cells, c-Macrophages

(3) a-Elastin fibres, b-Mast cells, c-Macrophages

(4) a- Collagen fibres, b-Macrophages, c-Mast cells

**82. The mushroom shaped gland in male cockroach is present in the**

(1) $4^{th} - 7^{th}$ abdominal segments

(2) $7^{th} - 8^{th}$ abdominal segments

(3) $6^{th} - 7^{th}$ abdominal segments

(4) $7^{th} - 8^{th}$ abdominal segments

**83. Cockroach is**

(1) Radially symmetrical, pseudocoelomate

(2) Bilaterally symmetrical, diploblastic and coelomate

(3) Bilaterally symmetrical, triploblastic and coelomate

(4) Bilaterally asymmetrical, coelomate

**84. Total number of ganglia in the nerve cord of cockroach is**

(1) Three thoracic and six abdominal

(2) Three thoracic and eight abdominal

(3) Two thoracic and six abdominal

(4) Three thoracic and three abdominal

**85. Head of cockroach is**

(1) Triangular

r/s lower fac

(2) Hypognathous

(3) Formed by fusion of six segments

(4) All of these

**86. Mark the incorrect statement for cockroach**

(1) Spermatophore is the bundle of sperms discharged during copulation

(2) Nymph undergoes ~~3-4~~ 13 moults to reach adult form

(3) Its development is paurometabolous

(4) Each ootheca has 14 to 16 eggs

**87. What is the function of the structure present at the junction of mid gut and hind gut in cockroach?**

(1) Storage of nitrogenous waste

(2) Removal of nitrogenous waste

(3) Absorption of nutrients

(4) Digestion of food

**88. Thin Malpighian tubules in cockroach are present at the junction of**

(1) Foregut and midgut

(2) Midgut and hepatic caecae

(3) Foregut and hindgut

(4) Midgut and hindgut

**89. Which of the following is correct w.r.t gastric caecae of cockroach?**

(1) 6-8 blind tubules at the junction of midgut and ileum to secrete digestive enzymes.

(2) 6-8 blind tubules at the junction of gizzard and midgut to secrete digestive enzymes

(3) 100-150 thin yellow tubules at the junction of foregut and midgut to eliminate wastes from haemolymph.

(4) 6-8 blind tubules at the junction of crop and gizzard which eliminate wastes from haemolymph

**90. Assertion : Eggs of cockroach are encased in capsules called ootheca**

**Reason : Ootheca is a dark reddish to blackish brown capsule**

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

Nuts

**91. Identify the correct path of passage of urine in female frog.**

(1) Kidney → urinogenital duct → cloaca → urinary → bladder

(2) Kidney → oviduct → urinary bladder → cloaca

(3) Kidney → ureter → ~~cloaca~~ → urinary bladder → cloaca

(4) Kidney → urinary bladder → ureter → cloaca

**92. The frogs have the ability to change its colour to hide them from their enemies. This "protective colouration" is called**

camouflage

(1) Hibernation

(2) Mimicry

(3) Aestivation

(4) Camouflage

**93. Select the correct answer**

**i. Body of a frog is divisible into head and trunk**

**ii. A neck and tail are sometimes absent**

(1) Only is i correct

(2) Only ii is correct

(3) Both are correct

(4) None are correct

94. Put the organs of the digestive system of the frog in its proper order

1. pharynx
2. Stomach
3. Mouth
4. intestine,
5. Oesophagus,
6. Cloaca
7. rectum

(1) 3, 1, 5, 2, 4, 7, 6

(2) 3, 1, 5, 2, 4, 6, 7

(3) 1, 3, 5, 2, 4, 7, 6

(4) 3, 1, 2, 5, 4, 7, 6

3, 1, 5, 2, 4, 7, 6

**95. Hindbrain of a "frog" consist of**

(1) Cerebellum only

(2) Cerebellum and medulla oblongata

(3) Medulla oblongata only

(4) Cerebrum and medulla oblongata

Thin walled ventral

**96. The ____ urinary bladder is present ____ to the rectum which also opens in the cloaca in the frog**

(1) Thick-walled, ventral

(2) Thin-walled, ventral

(3) Thick-walled, dorsal

(4) Thin-walled, dorsal

**97. The number of vasa efferntia that arises from testes in frog's male reproductive system is**

(1) 10 – 20

(2) 10 – 12

(3) 10 – 16

(4) 10 – 14

**98. Select the correct route for the passage of ova in female frogs**

(1) Ovaries → body cavity → oviducts → cloaca → exterior

(2) Ovaries → Bidder's canal → oviduct cloaca → exterior

(3) Ovaries → oviducts → body cavity → cloaca → exterior

(4) Ovaries → kidneys → cloaca exterior

**99. Assertion : Frog is poikilothermic**

**Reason : Frogs do not have constant body temperature**

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

**100. Select the correct answer about the frog**

**I. A mature female can lay ~~250 to 300~~ ova at a time.**

**II. Fertilisation is ~~internal~~.**

2500 – 3000

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(1) Only I is correct

(2) Only II is correct

(3) Both are correct

(4) None are correct

# THANK YOU

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